

Populace dynamics: an open, scored longitudinal layer for policy microsimulation

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Abstract

Dynamic microsimulation — the longitudinal aging of a person-level population through earnings, family structure, disability, mortality, and program participation — underpins retirement and social-insurance policy analysis, yet the benchmark United States models — DYNASIM, MINT, and CBOLT — are closed: internal to government, tied to restricted administrative records, or accessible only through institutional relationships. This design paper specifies an open alternative built as an extension of Populace, PolicyEngine’s open-source microdata stack, whose cross-sectional layer is the certified default United States microdata in PolicyEngine after outperforming its predecessor on held-out administrative targets. The design contributes four elements: a trajectory-weighted kernel in which multi-period calibration cannot silently destroy panel structure; a Dynamics operator that treats state transitions as conditional models, mixing deterministic demographic hazards with machine-learned earnings processes; an explicit domains-of-validity framework that refuses point forecasts where parameter uncertainty dominates — including the 75-year actuarial balance — and instead publishes sensitivity surfaces; and a scoring protocol under which every claim resolves against administrative publications, backtests with leakage control, or computes exactly from statute, with contributions merging only when they improve held-out scores. United States Social Security is the first validation domain; the layer itself is country-agnostic.

1 Introduction

Analysts who want to model taxes can run open, calibrated models directly: Tax-Calculator (Policy Simulation Library 2026) and PolicyEngine (PolicyEngine 2026) are openly callable on public data, alongside source-available models with restricted inputs (The Budget Lab at Yale 2026) and proprietary models used for outside-facing analysis (Tax Policy Center 2025; Institute on Taxation and Economic Policy 2025; Tax Foundation 2025; Penn Wharton Budget Model 2025). Analysts who want dynamic microsimulation — the longitudinal modeling that retirement and social-insurance policy requires — find the benchmark models closed: SSA projects retirement income and the distributional effects of policy proposals with MINT (Social Security Administration 2024), CBO produces its long-term Social Security projections with CBOLT (Congressional Budget Office 2018, 2024), the Urban Institute projects retirement income and long-term care with DYNASIM (Favreault et al. 2015; Urban Institute 2024), and Morningstar studies retirement adequacy with its Model of US Retirement Outcomes (Look and VanDerhei 2024). Outside users reach each only through an institutional relationship. The nearest open analogue, the Cato Social Security model (Chanwong 2026), simulates roughly 10,000 households from the 2007 CPS ASEC under the Social Security Administration’s assumptions and reports trust-fund metrics and reform scores, without published validation against administrative benchmarks.

This paper specifies the design of an open longitudinal layer — *Populace dynamics* — as an extension of Populace, PolicyEngine’s open-source microdata stack. It is a design paper: the cross-sectional

foundation is in production; this paper specifies the longitudinal layer, which the project builds in the open behind the scoring protocol of Section 5.

The design makes four contributions:

1. **A trajectory-weighted kernel with explicit alignment.** One weight per trajectory: multi-period calibration stacks constraint rows over the same weight vector, so that hitting cross-sectional totals in multiple periods cannot silently destroy panel structure — combined with event-selection alignment for period-by-period control, since weights alone cannot separate a correct life course from a correctly timed one (Section 4).
2. **Transitions as conditional models.** A single operator interface covers deterministic demographic hazards and machine-learned earnings processes, so candidate architectures compete under one evaluation standard (Section 4).
3. **Domains of validity as shipped metadata.** The model states which questions it will not answer with false precision — led by the 75-year actuarial balance, where input uncertainty dominates model fidelity — and publishes long-horizon results as sensitivity surfaces rather than point forecasts (Section 3).
4. **A scoring protocol in place of fidelity-only validation.** Claims resolve against administrative publications on an annual calendar, backtest against realized history with leakage control, or compute exactly from statute; contributions merge only when they improve held-out scores (Section 5).

United States Social Security is the first validation domain because benefit adequacy and reform incidence depend jointly on lifetime earnings, marriage and survivorship, disability, differential mortality, and claiming. The layer itself is country-agnostic and extends to other pension and benefit systems as PolicyEngine’s country coverage grows.

2 Terminology and scope

This paper uses “dynamic” in the microsimulation field’s standard sense, following Orcutt et al. (1961) and the tradition carried by DYNASIM, MINT, CBOLT, and SimPaths (Bronka et al. 2025): a longitudinal model that ages a person-level population through time (Li and O’Donoghue 2013). It does not mean “dynamic scoring” — the tax-policy usage denoting macroeconomic feedback in revenue estimation — and the design is not an overlapping-generations general-equilibrium model of the Auerbach–Kotlikoff kind, such as the Penn Wharton Budget Model operates (Penn Wharton Budget Model 2025). The model claims neither macroeconomic feedback nor equilibrium closure. Behavioral responses enter as labeled scenario inputs with documented ranges, not as point estimates carrying model authority.

3 Domains of validity

All models are wrong; a model earns its keep only where it improves predictions. The design therefore begins by bounding its own claims.

3.1 Parameter uncertainty dominates the long horizon

The 75-year actuarial balance of a pension system is a function of a small set of exogenous assumptions — fertility, mortality improvement, net immigration, real wage growth, interest rates — whose uncertainty dominates the microsimulation machinery that processes them. The public record makes this concrete for United States Social Security. The Trustees’ own low- and high-cost scenarios bracket a range of 75-year balances wider than the intermediate deficit itself (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds 2025). The Congressional Budget

Office’s long-term projections differ from the Trustees’ — chiefly for assumption reasons, though method differences such as CBO’s micro-founded projection of the taxable share of earnings also contribute to the divergence (Congressional Budget Office 2024). Assumption variance dominates the headline; model structure matters for distribution and the near term, which is the division of labor the output tiers encode. Successive Technical Panels convened by the Social Security Advisory Board have recommended revising fertility and mortality-improvement assumptions as realized values ran persistently outside the projected path (Technical Panel on Assumptions and Methods 2023). Two institutions with administrative data and decades of refinement disagree with each other and have both missed realized demographic trends; a better microsimulation does not repair that, because the variance lives in the inputs.

3.2 Three output tiers

The model sorts its outputs into tiers, each carrying the strongest claim it can support.

Tier 1: distributional analysis under fixed assumptions. Reform analysis is a difference — outcome under reform minus outcome under baseline, holding the population and assumption path fixed — and much of the unforecastable demographic uncertainty is common to both arms and cancels in the difference. The required ingredients — a calibrated joint distribution of lifetime earnings, family structure, and differential mortality, plus an exact rules engine — are the components the scoring protocol validates directly. The design labels rather than buries the slice of a reform delta that does not cancel: reform-induced claiming responses, and interactions between the reform and uncertain dynamics — the mortality gradient enters many deltas as a covariance with benefit position, not a level, so it does not difference out. Deltas evaluated past trust-fund depletion also require an explicit scheduled-versus-payable baseline convention, which the model states with every such output. For claiming, scenario ranges anchor to the quasi-experimental record on retirement-age responses (Mastrobuoni 2009; Behaghel and Blau 2012), and the model publishes them as a scenario library rather than embedding them as point estimates.

Tier 2: near-term components that resolve. Over roughly a ten-year horizon, mechanics rather than demographic extrapolation dominate outputs: beneficiary counts by type, average benefits, covered earnings and taxable payroll, claiming-age distributions, disability incidence. These resolve against administrative publications each year, and the protocol scores them (Section 5).

Tier 3: the long horizon as a sensitivity surface. The model publishes long-horizon outputs as surfaces over documented assumption ranges — how the balance, cohort replacement rates, or distributional outcomes move as fertility, mortality improvement, and immigration vary — never as point forecasts. The distinction is between computing and blessing: the model computes 75-year balances and depletion dates *conditional on named assumption paths*, including the Trustees’ intermediate path, so the numbers the policy debate runs on remain available as labeled conditional outputs; what the model declines is presenting any single path as its own forecast. Incumbent practice publishes the point estimate up front and the sensitivity analysis in an appendix; an open model can invert that and make the sensitivity the interface.

Every API response carries its tier, assumption path, and calibration history as metadata, so a downstream consumer — human or machine — weights the output by demonstrated reliability rather than by the producer’s reputation.

4 Architecture

4.1 The cross-sectional foundation

Populace builds a calibrated synthetic population entirely from primary-source government data — the Current Population Survey ASEC, the IRS Public Use File, the Survey of Consumer Finances, SIPP, CPS outgoing-rotation groups, MEPS, and the ACS — synthesizing missing variables with weight-aware conditional models and calibrating to administrative aggregates treated as uncertainty-weighted facts. In June 2026 it replaced PolicyEngine’s enhanced CPS as the certified default United States microdata in PolicyEngine, after a matched, symmetric-refit comparison on 41,314 households with a 739-target holdout (Table 1).

Table 1: Certification comparison from the Populace release manifest. The enhanced CPS wins more individual targets by small margins while its largest misses dominate the loss; the scoring protocol requires publishing the count that cuts against the headline.

Metric (lower is better)	Populace	enhanced CPS
Holdout loss (739 held-out targets)	0.038	0.317
Training loss	0.190	1.089
Full loss	0.228	1.405
Per-target wins	1,040	2,613 (51 ties)

4.2 Trajectory weights and population accounting

The longitudinal extension follows two kernel rules set in Populace’s charter. First, **one weight per trajectory**: multi-period calibration targets stack as (target, period) constraint rows over a single trajectory-level weight vector, so that calibrating cross-sections independently — which severs the trajectory-level consistency a panel exists to provide — is a kernel-level error rather than a modeling temptation. Second, **population is not closed**: trajectories carry entry and exit markers (birth, death, immigration, emigration), and a trajectory’s weight contributes to a period only while the person is present. Household and couple links are period-scoped, so family recomposition preserves per-period accounting identities; couples carry a shared unit weight derived from their trajectory weights so that spousal and survivor benefits have a well-defined representation.

Weights alone are one layer of alignment, not the whole answer: a weight cannot distinguish a correct life course from a correctly timed one, and reweighting trajectories to hit future cross-sectional cells risks selecting on entire correlated life courses. Period-by-period control therefore also operates through event selection — ranking individual transition probabilities and selecting the number of events an external control demands, the mechanism CBOLT and DYNASIM use — with trajectory weights reserved for base-year representation and slow-moving composition (Li and O’Donoghue 2013; Dekkers and Cumpston 2012). The kernel solves the stacked constraint system as uncertainty-weighted penalized least squares against target standard errors, so infeasible combinations resolve by SE-weighted compromise rather than silent failure, and projected demographic controls pin cohorts born after the base year rather than leaving them free.

4.3 The Dynamics operator

Dynamics is an operator from a population and a transition specification to a population with extended periods. A transition is a conditional distribution — the probability of next-period state given current state and covariates — which is the same interface Populace’s synthesis models already implement. The shipped baseline for earnings is a regime-gated, sequentially chained, weighted quantile-regression-forest imputer (Meinshausen 2006) whose zero-inflation gate doubles as a nonemployment model; richer

architectures (zero-inflated neural distribution models, normalizing flows) are candidates that must beat the baseline on held-out longitudinal moments to merge.

The design is hybrid. Where the evidence base is tabular — mortality from official life tables with published income gradients, fertility from vital statistics, marriage and divorce from ACS- and CPS-based rates (federal collection of detailed marriage and divorce statistics ended in the 1990s), disability incidence from program statistics — transitions are deterministic hazards, auditable row by row. Marriage requires a matching model, not only a hazard: spousal and survivor benefits depend on assortative matching over lifetime earnings, which the design treats as a first-class estimation target rather than an afterthought. The design reserves machine learning for processes with rich conditional structure, led by earnings dynamics — where the process is not stationary: volatility and mobility vary by cohort and period in administrative data (Sabelhaus and Song 2010; Kopczuk et al. 2010), so transition models carry cohort and period conditioning rather than pooling across decades. Chained one-period models also understate long-spell persistence unless spell structure enters the model explicitly, and backcasting is a distinct conditional object from forward simulation rather than the same operator reversed; both enter the evaluation as targets in their own right, disciplined by held-out panel moments such as higher-order earnings-change distributions (Guvenen et al. 2021).

The design states one measurement caveat rather than hiding it: the most demanding earnings-dynamics moments come from administrative records that public panels understate, and survey- and administrative-based estimates disagree on volatility levels and trends. Where the project cannot recompute a published administrative moment on held-out public data, matching it is calibration, not validation, and the scorecard labels it as such.

4.4 Rules and delivery

Statute is the deterministic slice of any policy forecast. Core retirement benefit formulas — average indexed monthly earnings, primary insurance amounts, actuarial adjustments — and benefit taxation compute exactly today through PolicyEngine’s rules engine via Populace’s rules-adapter protocol, vectorized over person-periods. Auxiliary, spousal, and survivor benefit formulas are explicit build items in the validation program: the current engine carries them as calibrated aggregates rather than person-level formulas, and the scorecard treats “computes exactly” as a per-rule status each formula earns, not a blanket claim.

The rules adapter is engine-agnostic. PolicyEngine-US implements it today; Axiom — an open project that encodes statute declaratively and compiles it to Rust — is the next adapter, and the performance headroom matters when benefit formulas run over person-periods across the full trajectory panel and many reform scenarios. In that architecture, PolicyEngine is a composition: Axiom supplies the rules, Populace supplies the population, and behavioral responses enter as the labeled scenario layer.

Data governance is a design requirement, not an afterthought. Estimating transition models on restricted-use panels and publishing the estimated parameters is settled practice — open models such as OG-USA, and DYNASIM itself, estimate on the PSID and publish what they learn. Releasing a synthetic *microdata* artifact informed by such panels is a stricter problem, because donor-based samplers can emit observed training values. The design answers it structurally: released records originate from Populace’s public-use cross-section, restricted panels train processes rather than donate records, samplers smooth or noise their draws so no verbatim donor value ships, and every release passes nearest-neighbor disclosure checks alongside its accuracy scorecard — and the project engages data producers directly where their terms of use warrant it. Uncertainty budgets therefore attach only to the components statute does not fix. The deliverable is a versioned artifact — a longitudinal population release with a manifest and scorecard, certified through the same path as the cross-sectional release — exposed through a Python library, a REST API, and a Model Context Protocol server so that AI agents can run baseline distributions and reform analyses with validity metadata attached.

5 Scoring and resolution

Validation by fidelity — does the model match published aggregates? — is necessary but weak: a model can reproduce the tables its authors fit it to. This project’s standard: a claim counts as validated when it improves prediction of something that later resolves. Five scoring surfaces implement it.

1. **Annually resolving components.** Beneficiary counts by type, average and aggregate benefits, covered earnings and taxable payroll, cost-of-living adjustments, disability incidence, and claiming-age distributions resolve against administrative publications each year. Every published forecast cell carries a resolution rule naming the exact table and vintage that settles it.
2. **Forecasting the forecasters.** Official projections revise every year; predicting the next revision of headline quantities resolves in months rather than decades and is decision-relevant to anyone who acts on the official number. This is partly forecasting an assumptions process — panels advise, committees adopt with a lag — so each cell pre-specifies the naive baseline it must beat (an assumption random walk plus mechanical data update).
3. **Retrodiction with leakage control.** Retrodiction builds the model from data vintages available at a historical date and scores it against realized outcomes. Populace’s versioned data registry pins vintages going forward; pre-registry history is a reconstruction problem — survey redesigns, revised administrative tables, re-released panels — so the protocol grades pre-registry backtests as pseudo-vintage, with a published log of deviations from true vintage, and no registry pins specification leakage: a “2005-vintage” model built today knows 2008 happened, and the protocol says so. Retrodictive calibration under the historical regime does not guarantee calibration under a new one, so backtests complement rather than substitute for live resolution.
4. **Statutory resolution.** Where an output is fixed by law, the rules engine computes it exactly, and enacted policy settles the corresponding conditional cells immediately.
5. **Held-out panel moments.** The protocol scores the population layer against moments it never fit: earnings-mobility matrices, autocorrelation and higher-order moments of earnings changes, cohort age-earnings profiles, and family-transition rates on held-out panel records.

Two governance rules complete the protocol. **Merge on score:** a contribution — a mortality module, a claiming model, an earnings architecture, from any contributor — merges if and only if it improves the population’s score on held-out facts, the rule Populace already applies to its cross-sectional layer. **Publication discipline:** misses publish with the same prominence as hits; superseded methods keep their historical scorecards; and stage gates in the development roadmap are pre-specified score thresholds, not narrative judgments.

Openness supplies most of the refereeing. Resolution rules are pre-registered, scores recompute from public data, and anyone who distrusts a published scorecard can rerun it — or fork the project and publish a rival scorecard. Two pieces sit beyond reproduction. First, restricted-data checks: the most demanding earnings-history moments live in linked administrative records that the public pipeline never touches, so the reader must trust whoever runs the comparison — author or outsider — and a validator without a stake in the result adds evidence where rerunning is impossible. Second, judgment: gate thresholds, disputed resolution rules, and the assumptions library carry discretion that pre-registration narrows but does not remove. The project defines its validation procedures and gate thresholds before the components they gate, and seats an advisory board to review them — and disputes under them — in public. Independent scoring across models is a decision only a third party can make; the design encourages it — published projections from the closed models give any such body a comparison set without model access — but does not depend on it.

6 First validation domain: U.S. Social Security

Social Security is the first domain because eligibility and benefits depend on the highest thirty-five years of indexed earnings, marital and survivorship histories, disability pathways, differential mortality,

and claiming timing — jointly. A layer that scores well here earns reuse in adjacent domains — Supplemental Security Income interactions, long-term care, and retirement adequacy, the last of which is *harder*, not easier: wealth projection challenged even administrative-data models (Favreault and Smith 2016), and a wealth and pension roadmap is future work, not an assumed extension — and in other countries’ pension systems.

The domain also makes the validity framework concrete. The canonical output of existing Social Security models — the 75-year balance and depletion date — is the quantity Section 3 declines to forecast. What the open layer offers instead is the combination the field lacks: tier-1 distributional incidence of reforms with reproducible assumptions; tier-2 near-term components with a public resolution record; and tier-3 sensitivity surfaces that make the assumption-dependence of long-horizon claims the interface rather than the appendix. No existing model, closed or open, publishes that combination. The closed benchmarks do publish validation and cohort tables; what outsiders cannot do is rerun the pipeline, vary its assumptions, or audit the intermediate states behind the published numbers.

7 Gate 1 in practice: the first pre-registered runs

The protocol of Section 5 stopped being a design in July 2026. This section reports what it produced: a locked gate, five registered and scored model runs, and the findings the failures purchased. Every number below recomputes from committed artifacts in the project repository; the run log lives in the repository’s pull requests and the candidate registry in [issue #42](#).

7.1 The lock

Gate 1 — earnings-history credibility — locked on 2026-07-05. Thresholds derive from committed noise-floor artifacts measured at the scale the protocol scores at (two disjoint 20%-of-persons samples of the PSID family earnings panel scored against each other), with every derivation stated as floor mean plus a named multiple of the floor’s seed standard deviation and enforced by a test: a floor rebuild that shifts any artifact fails continuous integration rather than silently orphaning the rationale. Before ratification the proposed thresholds went through three rounds of adversarial review, published in full on the ratification pull request. Round one found that the proposed numbers did not follow their own stated rule and were calibrated against a floor at five times deployment scale; round two found the same wrong-scale defect reintroduced on a second view and demonstrated that no window-2 statistic can catch a chained one-period model, forcing the persistence guard onto the battery’s long-horizon autocorrelation bands; round three verified the amendments and ratified. The merge of the ratification pull request is the lock event; the thresholds have not changed since, and any change requires a public amendment and a fresh review round.

7.2 Five runs, four failures, and what they isolated

Each candidate registers its complete specification — every modeling degree of freedom pinned — before its single scored run. The log autocorrelation ladder at 2, 4, and 10 years (locked bands 0.730 ± 0.05 , 0.657 ± 0.06 , 0.539 ± 0.07) tells most of the story:

candidate	2yr	4yr	10yr	verdict
chained weighted QRF (baseline)	0.726	0.573	0.333	fail
+ person effect as feature	0.726	0.688	0.649	fail

candidate	2yr	4yr	10yr	verdict
+	0.722	0.695	0.647	fail
persistence-aware decomposition				
structural	0.464	0.401	0.354	fail
three-component assembly				
donor splicing	0.779	0.704	0.616	fail
(single-donor)				
donor splicing	0.720	0.631	0.490	fail
(segments)				
Gaussian-copula	0.716	0.653	0.507	fail
rank dynamics				
empirical rank	0.692	0.548	0.381	fail
kernel				
k-NN rank	0.719	0.636	0.459	fail
bootstrap				
permanent-rank	0.791	0.733	0.670	fail
matching				
calibrated blend +	0.757	0.677	0.514	fail
Q0 regime				
inner-validated	0.773	0.695	0.510	fail
composition				
inner-validated	0.773	0.695	0.510	pass
composition (re-registered)				

The baseline failed exactly where the pre-lock review predicted a one-step chain must: window-2 geometry passed on every seed while the 10-year autocorrelation collapsed toward the Markov value. The second and third candidates added a latent person effect as a conditioning feature — first from a naive variance decomposition, then from a persistence-aware one that halved the drawn variance — and produced statistically identical ladders. That invariance is the program’s sharpest finding to date: a quantile forest rescales its response to a person-effect feature inversely to the feature’s scale, so conditional-draw generation transmits the raw within-panel person-mean variance share (0.647 on the training splits, inflated by transitory persistence over few biennial observations) no matter how the feature is constructed. No feature-side dial can set person-level variance. The fourth candidate set the variance structurally — assembling log earnings from an age profile, a person effect drawn at the decomposed permanent variance, a chained transitory component, and an observation-noise layer — and the variance landed as designed while the marginal distribution broke: parametric lognormal assembly inflated levels where the earlier candidates had inherited the data’s marginal from empirical conditional draws by construction.

The symmetry across the four failures defines the remaining design problem. Conditional-draw candidates preserve the marginal but cannot set person-level variance; the structural candidate set the variance and lost the marginal. The data demand both at once.

The generative track then ran the composition the taxonomy below suggests. A Gaussian-copula rank model — empirical quantile marginals, a calibrated permanent-plus-transitory latent, simulated-moment calibration — put the autocorrelation ladder inside its bands on every seed, the first generative candidate to do so, and its calibrated shares independently reproduced the variance decomposition estimated three candidates earlier; but Gaussian innovations churned earnings quintiles far too fast

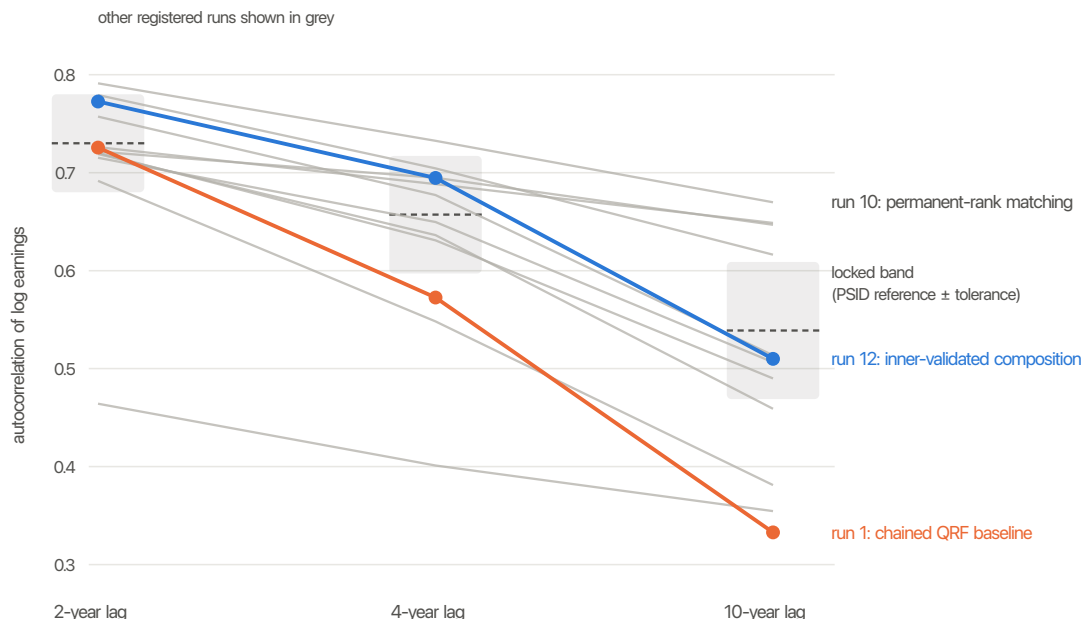


Figure 1: The autocorrelation ladder across the twelve distinct registered model runs. Shaded bands are the locked tolerances around the committed PSID reference at each horizon; run 13 re-registered the run-12 specification and reproduced it bit-exactly, so its curve coincides with run 12’s. Every value is computed from the committed run artifacts at figure-build time (`scripts/build_paper_figures.py`).

and the classifier read the synthetic joint easily. Replacing the Gaussian law with an empirical rank-transition kernel fixed the mobility matrix and halved the classifier gap while its one-step memory collapsed the ladder’s tail; adding two-step-plus-anchor memory through nearest-neighbor conditional draws brought the strongest scorecard of the sequence — the pairs-view classifier under its threshold on all five seeds, two seeds clearing the entire gate — with the window-3 classifier and the ten-year rung each missing on three seeds by thousandths. A final variant that matched donors on their estimated permanent rank rather than their observed anchor converted anchor noise into permanent signal and overshot every rung: the tenth registration, and the second time the program measured the same lesson from the opposite side.

7.3 How existing models solve the joint problem

Every serious earnings-history model confronts the same triple — the cross-sectional marginal, person-level persistence, and measurement noise — and the field has four working answers, each buying one thing at a stated price.

Reuse observed careers. MINT splices segments of donor workers’ administrative earnings records onto targets matched on demographics and recent earnings, adding a person-specific fixed effect in its regression projections (Social Security Administration 2024). Reuse dissolves the joint problem rather than solving it: the marginal is exact because the values are real, and person-level dependence is exact because whole careers come from one person. The price is donor support — no pattern appears that no donor exhibited, cohort drift needs ad hoc adjustment, and conditioning is only as rich as the match cells.

Generate parametrically, then align. DYNASIM carries an individual-specific error term in its

registered run	geometry seeds passed (of 5)	battery seeds passed (of 5)	pooled Q0	gate 1
1. chained weighted QRF (baseline)	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
2. + person effect as feature	○ ● ● ● ○	○ ○ ○ ○ ○	not scored	× fail
3. + persistence-aware decomposition	● ● ● ○ ○	○ ○ ○ ○ ○	not scored	× fail
4. structural three-component assembly	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
5. donor splicing (single-donor)	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
6. donor splicing (segments)	○ ○ ○ ○ ○	● ● ● ● ●	not scored	× fail
7. Gaussian-copula rank dynamics	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
8. empirical rank kernel	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
9. k-NN rank bootstrap	○ ● ○ ● ○	● ● ○ ○ ○	not scored	× fail
10. permanent-rank matching	○ ○ ○ ○ ○	○ ○ ○ ○ ○	not scored	× fail
11. calibrated blend + Q0 regime	○ ○ ○ ○ ○	○ ○ ● ● ○	×	× fail
12. inner-validated composition	○ ○ ● ● ●	○ ● ● ● ●	✓	× fail
13. inner-validated composition (re-registered)	● ● ● ● ●	○ ● ● ● ●	✓	✓ pass

● seed passed ○ seed failed
 gate 1 requires $\geq 4/5$ seeds on geometry AND $\geq 4/5$ on battery;
 runs 11–13 were also scored on the amended benefit-space block and pooled Q0; run 13's pairs-view classifier is gated by the amendment-2 rule (mean over 20 pre-registered seeds ≤ 0.53 with a per-seed cap of 0.554) rather than per-seed at 0.53.

Figure 2: Per-seed gate conjunction for the thirteen registered runs, from the committed run artifacts. Each cell shows the five locked holdout seeds; gate 1 requires at least four of five on geometry and on battery jointly. Runs 11–13 were additionally scored on the amended benefit-space block and the pooled zero-anchor gate, and run 13's pairs-view classifier is gated by the ratified amendment-2 rule — the mean over twenty pre-registered seeds at the unchanged 0.53 line with a 0.554 per-seed cap — rather than per-seed.

earnings equations (Favreault et al. 2015; Urban Institute 2024); CBOLT decomposes each worker’s earnings into a permanent shock — the long-run gap from the group mean — and a transitory shock, estimated on administrative panel records (Congressional Budget Office 2018). Both then align: simulated outputs are recalibrated, typically rank-preservingly, to external distributional and aggregate targets (Li and O’Donoghue 2013). Alignment is the field’s standing admission that generated marginals drift; it re-imposes them after the fact and guarantees the published tables. The documented cost is that alignment can distort the relationships between variables and hits its targets whether or not the underlying process is right (Li and O’Donoghue 2013) — the class of silent correction this project’s gate exists to expose rather than absorb: alignment would have masked the fourth candidate’s marginal failure entirely.

Fit the simulator, not the one-step step. The administrative-data literature estimates rich nonlinear processes by simulated method of moments — parameters chosen so that simulated trajectories reproduce the target moments jointly (Güvenen et al. 2021). This answers the composition problem directly: one-step conditionals estimated from noisy data need not compose into correct long-horizon dynamics, so the generator is scored as a whole, at the horizon that matters. The price is a parametric process and total dependence on the chosen moment set.

Put the dynamics in rank space. Nonlinear panel frameworks place persistence on a latent quantile position and read earnings off the empirical quantile function (Arellano et al. 2017). The marginal becomes untouchable by construction and all modeling effort concentrates on the rank process, where the permanent–transitory structure lives — at the price of heavier estimation machinery than any off-the-shelf learner supplies.

The fifth run tested the first strategy as a benchmark, in its simplest registered form: one donor per target, whole careers spliced by age with a level adjustment at the anchor. It failed — informatively. The marginal came through exactly as reuse promises (every distributional and tail metric passed comfortably), but the omnibus classifier still separated spliced from held-out trajectories on every seed, and the battery found whole-career reuse too persistent: mobility, exit rates, and zero-spell lengths all sit outside their bands, and the autocorrelation ladder overshoots at every horizon. The run’s own diagnostics name the artifacts — donor and target cohorts occupy offset points of the biennial age grid, so half the spliced values come from an adjacent age; scaling a whole career to a noisy anchor observation converts that noise into a permanent component; and one donor per career is strictly more persistent than the segment splicing MINT actually performs. The result bounds naive splicing rather than refuting the strategy. The registered segment variant — three-period segments from multiple donors, spliced by calendar period and level-adjusted at each segment boundary — then became the first candidate to clear the entire battery: all five seeds pass every dynamics tolerance, with the autocorrelation ladder, mobility, exit rates, and spell lengths inside their bands at once. What remains is a single metric family: the omnibus classifier still separates spliced from held-out trajectories by 0.017 and 0.027 above its two thresholds, with every other geometry metric passing on every seed. Six runs in, the gate has narrowed from “the dynamics are wrong” to “a classifier can still tell,” which is the precise question the generative track now has to answer. The generative track’s next design composes the third and fourth strategies with the project’s existing machinery: empirical quantile marginals, persistent dynamics in rank space, and innovation dispersion calibrated so iterated — not one-step — dynamics match the data’s ladder (Arellano et al. 2017; Güvenen et al. 2021).

Two auxiliary results harden the target. Excluding every PSID observation carrying an income-assignment flag (8.7% of positive-earnings person-periods overall, rising to 14–18% in the 2020 and 2022 waves) moves the 10-year reference by +0.012 — survey imputation does not explain the gap the candidates must close. And the statutory-resolution surface of Section 5 produced its first artifact: the project’s Python benefit oracle and the Axiom rules engine compute identical primary insurance amounts — 240 of 240 synthetic careers exact to the cent, both code paths pinned by revision — after the cross-engine comparison surfaced and fixed a statutory off-by-one in the elapsed-year count and a

rounding-direction error in a test fixture that no single-engine test had caught.

7.4 Recalibrating the gate toward decision relevance

Nine runs in, the maintainer asked the governance question the protocol exists to make askable: is the bar too high? The program answered with analyses rather than judgment. Classifier forensics showed the two best candidates' identical residuals were distinct defects, not a shared wall. A downstream-relevance analysis then pushed generated and real careers through the project's own statutory benefit calculator: the best candidate's benefit distributions were statistically indistinguishable from real ones — the distributional distance inside the real-versus-real noise floor, central gaps under two percent against a five-percent criterion — while the same analysis exposed a defect no locked metric had isolated: careers generated for people observed with zero earnings at their anchor overstated their benefits by nine percent, exactly the population a progressive benefit formula weights most. The locked gate was strict on an axis that does not matter for benefits and silent on one that does.

The response was the contract's own amendment mechanism, exercised in full for the first time: a proposal committed as an inert object changing nothing; a fresh adversarial referee round (which caught a misattributed criterion citation and forced disclosure that the proposed demotion flips four historical geometry verdicts — none of which changes an overall outcome); fixes; a verification pass; maintainer ratification by merge; and a follow-up flipping the ratified content into the locked block. The amended gate demotes the benefit-immaterial window-3 classifier to reported status and adds a gated benefit-space block — distributional and decile bands at the five-percent criterion, a distance bound derived from a committed real-versus-real anchor, and a pooled band on the zero-anchor subgroup where reality measures under three percent and the best candidate nine. Recalibration added strictness where the evidence says it matters: the best candidate still fails the amended gate, and reality still passes it.

7.5 Nested validation and the forecast discipline

Two practices matured in the runs that followed the amendment. The eleventh candidate composed the two best-understood mechanisms and failed through two couplings its registration had not foreseen — including a swing of the zero-anchor benefit gap from +9 to −18 percent through an interaction between its memory coordinate and the very subgroup it was fixing. The response was methodological rather than another guess: the one-shot rule protects the outer holdout, so model development may iterate freely on inner splits carved from each seed's training complement. An inner-validation harness now mirrors the amended gate at inner scale, and a design sweep raced the candidate mechanisms head to head on it — establishing with no outer-holdout contact that the zero-anchor participation refit alone closes the benefit gap, that the drawn persistent-state coordinate fixes the dynamics battery completely while wrecking the joint completely, and that one composition stood near-clearing everything at once. The twelfth candidate froze that composition, every constant selected by nested validation.

Each of the last two runs also carried a pre-registered forecast — component probabilities logged on the public registry before the run, graded against the outcome after. The twelfth run graded almost exactly: forecast 0.42 with modal failure named as the pairs-view classifier one seed short, and that is what happened — the battery passed (the ten-year rung in-band on every seed for the first time), the zero-anchor gap closed to +0.04 percent, the per-seed benefit metrics passed everywhere, and two seeds clipped the pairs-view classifier by 0.0015 and 0.0030. Twelve registered runs in, the program's entire remaining distance to its own gate is a classifier residual of thousandths on one view.

A twenty-seed extension of that measurement — reported, not gated — placed the residual against the matched real-vs-real floor (Figure 3). The twenty-seed mean sits 0.0066 below the locked 0.53 line; the five locked gate seeds alone average 0.0021 below it, and the two seeds that clip the line are the two largest candidate scores across all twenty. Whether that pattern is seed noise or a seed-set property

became the second amendment proposal, and the adversarial referee round earned its keep: the referee found the locked seeds non-exchangeable with the fresh ones in exactly the damaging direction (a random five-subset has a mean that high with probability 0.009), found the proposal's headline margin describing the twenty-seed mean where the gate scored the locked-five mean — one standard error below the line, not five — and named the self-rescue: unlike the first amendment, whose trigger still failed after ratification, this one's triggering candidate would have flipped to a pass.

7.6 The first pass

The reworked amendment survived a verification round and was ratified as an estimator change that refuses its own trigger. The pairs-view classifier now gates on the mean over twenty pre-registered seeds at the unchanged 0.53 line — an operating characteristic *stricter* above the line than the per-seed rule it replaces — with a per-seed catastrophe cap re-derived from a classifier-version-matched floor, and two standing rules: no candidate's committed verdict changes under a rule proposed after its own run, and floor derivation and candidate scoring must share a classifier version. Run 12's verdict stands as a fail permanently.

The thirteenth registered run was therefore a fresh registration of the identical specification, with the protocol's plainest disclosure yet: its pairs-classifier outcome was already public before the run, so the registered forecast was 0.97 with the residual entirely on execution error. The run reproduced every committed baseline bit-exactly — all twenty pairs scores, the locked-seed battery, benefit, and geometry blocks, to the last digit — and passed every block of the amended gate: geometry five of five, battery four of five, pooled zero-anchor gap +0.04 percent, twenty-seed classifier mean 0.5234 against 0.53 with a maximum seed at 0.5330 against the 0.554 cap (Figure 2). The first pass of the program arrived, in other words, not from a better model but from a better-measured gate — and the record distinguishes those two things explicitly, which is the point of keeping one.

A registered reform-delta diagnostic followed the pass (reported, not gated): two opposite-incidence mechanical reforms — the first PIA factor raised from 90 to 95 percent, and the taxable maximum removed from the AIME step — computed on real versus generated histories under the locked holdout protocol, each gap measured against a real-vs-real half-split floor. The aggregates a reform score leads with land inside the floor on both reforms: mean benefit change (gap \$0.29 against a \$0.48 floor bottom-loaded; \$4.02 against \$7.77 top-loaded), winners share, and the zero-anchor subgroup. The fine incidence curve is not fully reproduced: the bottom-loaded reform misses only the concave first-bend decile (1.9 times its floor, about one percent of the flat \$51 monthly delta), while cap removal shifts roughly a quarter of the top compared decile's gains downward — real +\$75.76 a month at the ninth decile against generated +\$55.33 — with six of seven compared deciles individually outside their floors. The named mechanism is that the generator under-concentrates cap-riding careers, diluting the extreme top of the earnings distribution, consistent with the run-12 microtexture forensics; it is the program's next target, found by the protocol's own diagnostics rather than by a downstream user.

7.7 What failure buys

Nineteen registered runs across two gates — thirteen at gate 1, six at gate 2 — eighteen published failures against a single pass under a rule that refused to rescue its own trigger, two refereed gate-1 amendments and a second gate locked through its own adversarial round, five external-anchor replications reported at the same standing as any gated run, and more than ten graded forecasts and directional calls is the protocol working as designed. Each failure published with the same prominence as a pass would, each narrowed the design space with a finding that transfers beyond this project, and none required trusting the authors: the registrations predate the runs, the artifacts recompute, and the thresholds cannot drift to accommodate a result — when the gate itself was recalibrated, it moved only through a public, refereed, ratified amendment whose every verdict change is disclosed in the

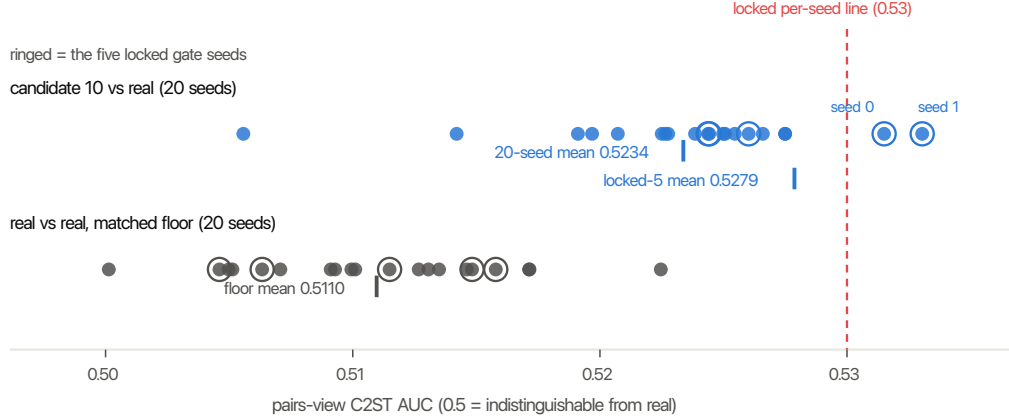


Figure 3: The twenty-seed extension of the run-12 pairs-view classifier measurement (reported, not gated), from the committed diagnostics artifact. Blue: candidate 10 scored against real holdouts; grey: the matched real-vs-real floor. Ringed points are the five locked gate seeds; both the twenty-seed and locked-five candidate means are marked.

contract’s own history. The contrast with validation by fidelity — where a model meets the tables it was fit to and the first hard test arrives after adoption — is the point.

8 The demographic layer and the replication anchors

The tally above compresses two developments the same protocol governed once gate 1 produced its first pass: a second stage gate, opened and locked on the family-transition layer, and the reform-scoring surface, tested against external anchors on real microdata. Neither is finished; both are on the public record with the discipline the earnings-history gate established.

8.1 The second gate’s lock ceremony

Gate 2 — family and benefit outputs — governs the demographic transitions survivor, spousal, and caregiver reforms score on: first-marriage, divorce, widowhood, and remarriage hazards, cohort nuptiality, fertility, and the dissolved-state stock shares eligibility rides on. It locked on 2026-07-08, through the ceremony gate 1 established: draft floors, an adversarial round, fixes, verification, and ratification by merge.

The draft built the reference moments and a person-disjoint half-split noise floor on five split seeds, with each per-cell tolerance the floor mean plus four standard deviations across forty gate-eligible cells. The adversarial round returned *amend before lock*, and its objection was quantitative: on five seeds the floor’s standard deviation is a five-draw estimate of a half-normal, so the committed tolerances realized anywhere from 0.9 to 5.6 times their own measured noise, and a faithful candidate — one drawing from the reference process itself — cleared the four-of-five gate with probability 0.023. The referee also showed that the draft’s floor scale, pass statistic, and seed rule described three different experiments, and that a verbatim copy of a training half passed at the noise floor, which no moment gate can prevent — the memorization defense is procedural (registration, holdout exclusion, and the no-self-rescue rule), not a property of the cell set.

The fixes rebuilt the floor on one hundred split seeds, stabilizing the estimator to roughly 3.2 times each cell’s own noise; added a power cap that gates a cell only when its stabilized tolerance is at

most $\ln(1.5)$ — a 1.5-times rate error — demoting under-powered per-age cells to report-only and recovering their coverage through pre-registered aggregates; and added the sequence and stock statistics a banded-marginal gate misses: origin-split remarriage, cohort ever-married-by-40, and dissolved-state stock shares by age and sex. Under the same rule on the rebuilt floor, the faithful-candidate operating characteristic is 0.969 over the resulting forty-six gated cells. Verification confirmed the amendments, and maintainer ratification by merge was the lock event. Every figure recomputes from the committed floor artifact (`runs/gate2_floors_v2.json`).

The gate’s external anchors are shape reports, not level gates, and the ceremony made the reason explicit. The raw recent PSID marriage and divorce rates run about 2.4 times the national vital-statistics rates — but the PSID counts persons transitioning where the vital-statistics series counts couples (a factor of two exactly) against a person-year denominator for the age-15-plus population (a further 1.22). Attributing those two concept factors leaves residuals of 1.036 for marriage and 0.989 for divorce: the anchor is a near-bullseye once the population concepts are aligned, and the period-matched fertility series sits at a median PSID/NCHS ratio of 0.93.

8.2 The gate-2 candidate ladder

Six candidates have run against the locked family-transition gate, each registered on the campaign registry ([issue #42](#)) before its single scored run and graded against a pre-registered forecast after. The ladder narrows the failure without yet closing it. The base composition — stratified empirical hazards with a mortality-composed widowhood component — fails all five seeds across fifteen distinct cells. Adding an age-by-sex first-marriage interaction and a decade-period widowhood mortality makes the composed widowhood backfire, widening the failure to nineteen cells. Pooled-rate shrinkage un-explodes that cascade back to eight; a parametric per-sex mortality trend then fixes the 65–74 female widowed stock but lets the residual go diffuse. Replacing the trend’s source — PSID’s own male mortality slope was unstable — with an external NCHS life-table trend, and adding single-year-kernel fertility and an origin-split remarriage table, narrows the failure to six cells. The sharpest fix is the sixth: source-aligning the spouse-death level to the surviving spouse’s own marriage-history widowhood incidence removes a sex-asymmetric wedge in the earlier death-record level — which understated female widowhood by 1.9 to 2.8 times while overstating male — lifting the 75-plus female widow stock from one seed to four and clearing one full seed. Its residuals are the untargeted male lifetime-marriage sequence cell, two seeds over its 0.047 tolerance; a three-cell miss on a third seed; and the fertility clip inherited from candidate 5 on a fourth — one of the five seeds clears, the best result to date.

Table 2: The gate-2 candidate ladder, from the committed run artifacts (`runs/gate2_hazard_v1.json` through `v6.json`). “Distinct failing cells” counts gated cells missing on at least one of the five locked seeds; the gate requires at least four of five seeds with every one of the forty-six gated cells inside its locked tolerance.

candidate	distinct failing cells	seeds passing	headline lesson
1 — stratified hazards, composed widowhood	15	0 / 5	the base composition; young first-marriage and female widowed-stock cells drift
2 — + age×sex first marriage, period widowhood mortality	19	0 / 5	the composed widowhood backfires — the widest failure
3 — + pooled-rate mortality shrinkage	8	0 / 5	shrinkage un-explodes the cascade; the widowed stock persists

candidate	distinct failing cells	seeds passing	headline lesson
4 — + parametric per-sex mortality trend	8	0 / 5	fixes the 65–74 female widowed stock; the residual goes diffuse
5 — + external NCHS mortality trend, kernel fertility	6	0 / 5	an external trend replaces PSID’s unstable male slope
6 — + source-aligned spouse-death level	5	1 / 5	removes the sex-asymmetric widowhood wedge (1.9–2.8×); the best result

8.3 The replication anchors

While the generative track worked the demographic gate, the reform-scoring surface — the statutory AIME/PIA chain and the survivor, spousal, and caregiver benefit plumbing, run on real PSID careers — was tested against five external DYNASIM anchors. Each is reported, not gated: registered on the campaign registry before the run and published regardless of outcome. Four score real careers directly; the price-indexing anchor additionally routes generated careers through the same calculator.

Table 3: The five external-anchor replications, reported not gated, from the committed diagnostic artifacts; each registered on [issue #42](#) before its run.

provision (anchor)	achieved replication	artifact / registration
progressive price indexing — Mermin (2005)	the price-indexing scalar lands at 66.73% of scheduled against DYNASIM’s 67.8%; the progressive-price-indexing quintile pattern reproduces — real and generated careers decline together from 100% to 84% of scheduled across quintiles, matching DYNASIM’s monotone gradient — with real-versus-generated gaps inside the floor at four of five quintiles	replication_ppi_mermin_v1.json; #42 c4907444903
normal-retirement-age increase to 70 — Mermin (2005)	79.75% of scheduled overall, cross-quintile spread 0.045pp, each quintile within 0.03–0.33pp of Mermin’s 79.4–79.9% row	replication_mermin_rows_v1.json; #42 c4911609804
COLA reduced 0.4pp — Mermin (2005)	99.2% of scheduled at ages 62–67 (anchor 98.9%) and 93.0% at 80–85 (anchor 92.4%)	replication_mermin_rows_v1.json; #42 c4911609804

provision (anchor)	achieved replication	artifact / registration
earnings sharing — Favreault and Steuerle (2007)	the best-populated cell — married women, the large gainers — lands on DYNASIM: gain 20% 49 vs 44, gain 5% 63 vs 60, lose 5% 24 vs 22; all four registered directional calls hold	<code>replication_r7_sharing_v1.json</code> ; #42 c4911171806
caregiver credit — Smith et al. (2020)	all four candidate plans place 58–71% of aggregate gains in the bottom lifetime-earnings quintile, in or above the anchor’s 52–62% band; the concentration and reach rankings match (Spearman 0.8)	<code>replication_caregiver_v1.json</code> ; #42 c4911453454

The discipline that makes those comparisons legible is that every level difference from the DYNASIM projections is named, not absorbed. The populations differ — observed PSID retirees eligible 2005–2019, born 1943–57, against DYNASIM’s projected 1960–80 cohorts evaluated in 2049 — as do the windows (a biennial PIA-proxy convention, the national average wage projected forward to each indexing year) and the earnings concept (the earnings-sharing exercise scores real couples with both spouses computable, a shared-versus-individual distinction the incidence turns on). Where the older PSID cohorts hold more single-earner couples, married men lose harder than DYNASIM’s 2049 projection, and the direction, not the level, is what the test certifies. The same discipline caught an error in an anchor: Favreault and Steuerle’s Table 3 carries a package-1c married-women column summing to 109.1, a 9.1 in a “no change” cell where every comparable earnings-sharing cell reads 0.0 — a printed typo, flagged and carried verbatim, feeding no scored result.

Two things are true at once, and the record keeps them distinct. Distributional scoring on validated real microdata is anchor-grade across the cataloged provision classes — price indexing, retirement-age and cost-of-living reductions, earnings sharing, and caregiver credits — each reproducing its DYNASIM incidence pattern within the noise floor or a named delta. The generated-population demographic track has not reached its own gate: six registered candidates in, the strongest clears one of five seeds. The paper reports both because the protocol scores both, and a reader who wants to know which claims are load-bearing today can read it off the record rather than the prose.

9 Related work

Dynamic microsimulation originates with Orcutt et al. (1961); Li and O’Donoghue (2013) survey the field’s alignment and calibration practice. This design borrows the central structural lesson of DYNASIM (Favreault et al. 2015; Urban Institute 2024), MINT (Social Security Administration 2024), and CBOLT (Congressional Budget Office 2018) — an annual state engine with family links and external alignment — while departing on openness, trajectory-level weighting, and scoring; Dekkers and Cumpston (2012) treat the weighting question directly. The Cato model (Chanwong 2026) is the nearest open system.

The pattern repeats internationally — Pensim2 at the United Kingdom’s Department for Work and Pensions, MOSART at Statistics Norway, MIDAS at Belgium’s Federal Planning Bureau (Li and O’Donoghue 2013) — with exceptions. INSEE has published the source of Destinie 2 (Blanchet et al. 2010), the pension model behind France’s official projection exercises (github.com/InseeFr/Destinie-2). SimPaths, from the University of Essex’s Centre for Microsimulation and Policy Analysis, is an open-source life-course model estimated for the United Kingdom and being adapted to several other European

countries (Bronka et al. 2025). And open frameworks exist without open populations: LIAM2 — built at the same Federal Planning Bureau that runs MIDAS (Menten et al. 2014) — and OpenM++, an open reimplement of Statistics Canada’s Modgen platform (OpenM++ development team 2026), supply generic simulation engines but ship no calibrated population and no rules stack. On OpenM++, WIFO’s microWELT models welfare transfers comparatively across Austria, Spain, Finland, and the United Kingdom, with a United States variant for labor-force projection (Spielauer et al. 2020) — one open engine carrying a multi-country dynamic model. None of these combines an open codebase with a certified calibrated microdata baseline and a public scoring protocol under which claims resolve. SimPaths is closest on openness; the difference here is inheritance — a certified cross-sectional population, one rules platform across countries, and credibility staked on resolved scores rather than publication.

On the measurement side, administrative-data studies of earnings dynamics discipline the earnings process: lifecycle moments (Guvenen et al. 2021), long-run mobility (Kopczuk et al. 2010), non-stationary volatility (Sabelhaus and Song 2010), and the attenuating link between current and lifetime earnings (Haider and Solon 2006) — with nonlinear panel frameworks (Arellano et al. 2017) the academic cousin of the quantile machinery used here. The quasi-experimental record on claiming responses (Mastrobuoni 2009; Behaghel and Blau 2012) anchors the behavioral scenario library, and the assumption-failure record compiled by successive Technical Panels (Technical Panel on Assumptions and Methods 2023) motivates the domains-of-validity framework.

10 Status and roadmap

The cross-sectional foundation is in production. Populace’s charter specifies the longitudinal kernel rules. This paper is the project’s front door; the supplementary design appendices — operational chapters on earnings-history construction, family and auxiliary benefits, disability and claiming, mortality and projection drift, calibration targets, the Social Security validation program, and a source-based DYNASIM dossier — live in the project repository at github.com/PolicyEngine/populace-dynamics. Development proceeds through stage gates defined as score thresholds — earnings-history credibility first, family and benefit outputs second, forward projection third, productization last — and every gate’s evidence publishes whether it passes or fails.

The layer is open source under the MIT license. The project invites corrections, particularly to the benchmark characterizations, and contributions merge on the same standard as the authors’: improve the held-out score.

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